

1 Initial Formulation

Beginning with the one dimensional RTE

$$\frac{1}{c} \frac{\partial}{\partial t} \psi(\nu, t, x) + \Omega_x \left(\frac{\partial}{\partial x} \psi(\nu, t, x) \right) = 4\pi \sigma_a B - \sigma_a \psi(\nu, t, x). \quad (1)$$

With the material energy equation

$$C_v \frac{\partial T(t)}{\partial t} = \int_0^\infty d\nu (\sigma_a \phi(\nu, x, t) - 4\pi \sigma_a B). \quad (2)$$

Discretizing the derivatives on the RTE gives

$$\frac{1}{c} \frac{\psi_i^{n+1} - \psi_i^n}{\Delta t} + \Omega_x \left(\frac{\psi_{i+1/2}^{n+1} - \psi_{i-1/2}^{n+1}}{\Delta x} \right) = \sigma_a B - \sigma_a \psi_i^{n+1}. \quad (3)$$

If we move the last time step to the rhs we obtain

$$\left(\frac{1}{c\Delta t} + \sigma_a \right) \psi_i^{n+1} + \Omega_x \left(\frac{\psi_{i+1/2}^{n+1} - \psi_{i-1/2}^{n+1}}{\Delta x} \right) = \sigma_a B + \frac{1}{c\Delta t} \psi_i^n. \quad (4)$$

For ease of use we let

$$Q^* = \sigma_a B + \frac{1}{c\Delta t} \psi_i^n \quad (5)$$

$$\sigma^* = \left(\frac{1}{c\Delta t} + \sigma_a \right). \quad (6)$$

Assuming Q is time invariant for now, and that we are in a 1-D space, we have obtained the steady state eq

$$\mu_l \left(\frac{\psi_{i+1/2}^l - \psi_{i-1/2}^l}{\Delta x} \right) + \sigma^* \psi_i^l = Q^*. \quad (7)$$

We can now add an upwind scheme for the spatial term

$$\frac{d\psi_i}{dx} = \left(\frac{\psi_i^l - \psi_{i-1}^l}{\Delta x} \right), \quad \mu_i > 0 \quad (8)$$

$$\frac{d\psi_i}{dx} = \left(\frac{\psi_{i+1}^l - \psi_i^l}{\Delta x} \right), \quad \mu_i < 0 \quad (9)$$

The next term update will for $\mu_l > 0$ will be

$$\frac{\mu_l \psi_i^l}{\Delta x} - \frac{\mu_l \psi_{i-1}^l}{\Delta x} + \sigma_a \psi_i^l = Q^*. \quad (10)$$

This makes the update

$$\psi_i^l = \frac{Q^* + \frac{\mu_l \psi_{i-1}^l}{\Delta x}}{\frac{\mu_l}{\Delta x} + \sigma^*} \quad (11)$$

For $\mu_l < 0$

$$\psi_i^l = \frac{Q^* + \frac{\mu_l \psi_{i+1}^l}{\Delta x}}{\frac{\mu_l}{\Delta x} + \sigma^*} \quad (12)$$

Going back to our full equation and discretizing angle we have

$$\mu_l \left(\frac{\psi_{i+1/2}^{n+1,l} - \psi_{i-1/2}^{n+1,l}}{\Delta x} \right) + \sigma^* \psi_i^{n+1,l} = Q^{*l}. \quad (13)$$

Where

$$Q^{*l} = \sigma_a^l B^l + \frac{1}{c \Delta t} \psi_i^{n,l}, \quad (14)$$

Where $\psi_i^{n,l}$ is from the last time step. We can make the rhs lag to be fully a function of the last time step by making $B(T) = B(T^n)$ making the rhs known. From here we must consider the material equation

$$C_v \frac{\partial T}{\partial t} = \int_0^\infty d\nu \sigma_a \phi(x, \nu, t) - 4\pi \sigma_p a c T^4. \quad (15)$$

Discretizing the time steps gives

$$C_v^{n+1} \left(\frac{T^{n+1} - T^n}{\Delta t} \right) = \int_0^\infty d\nu \sigma_a \phi(x, \nu, t)^{n+1} - 4\pi \sigma_p a c (T^4)^{n+1}. \quad (16)$$

We also have the trick that

$$T_{n+1}^4 \approx T_n^4 + 4T_n^3(T^{n+1} - T^n) \quad (17)$$

After making a gray code, we plot a few test problems from this formulation. The two test problems we do are Reed's problem and the gray Marshak wave problem. We do both with a reflective boundary condition at $x = 0$ and a true zero right hand boundary. For both the simple source is used setting Q^* to a constant as opposed to utilizing the coupling with the Material equation.

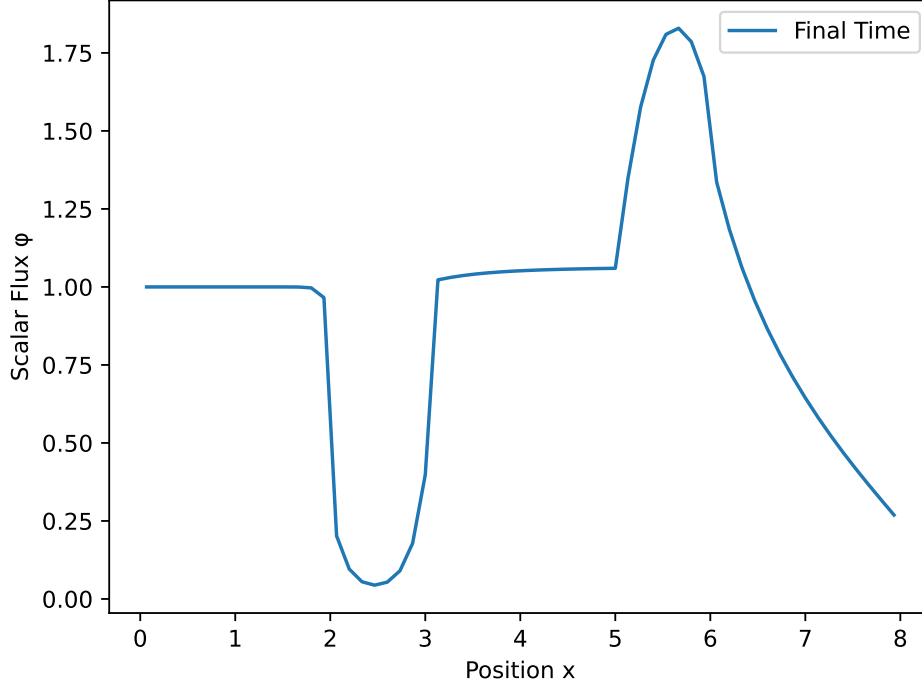


Figure 1: Reed's Problem

2 MultiGroup Formulation

Now that we have the Gray form, we can begin the multigroup formulation. Instead of integrating to find the group average of the discretized equation, we can once again begin from the one dimensional RTE.

$$\frac{1}{c} \frac{\partial}{\partial t} \psi(\nu, t, x) + \Omega_x \left(\frac{\partial}{\partial x} \psi(\nu, t, x) \right) = 4\pi\sigma_a B - \sigma_a \psi(\nu, t, x). \quad (18)$$

With the material energy equation

$$C_v \frac{\partial T(t)}{\partial t} = \int_0^\infty d\nu (\sigma_a \phi(\nu, x, t) - 4\pi\sigma_a B). \quad (19)$$

We proceed by integrating over the frequency group bounds to get the group average

$$\int_{\nu_{i-1/2}}^{\nu_{i+1/2}} d\nu \left(\frac{1}{c} \frac{\partial}{\partial t} \psi(\nu, t, x) + \Omega_x \left(\frac{\partial}{\partial x} \psi(\nu, t, x) \right) = 4\pi\sigma_a B - \sigma_a \psi(\nu, t, x) \right). \quad (20)$$

Luckily, we may define

$$\bar{\psi}_g(t, x) \equiv \int_{\nu_{i-1/2}}^{\nu_{i+1/2}} \psi(\nu, t, x) d\nu \quad (21)$$

$$\bar{\sigma}_{\text{ag}}(x, t) \equiv \int_{\nu_{i-1/2}}^{\nu_{i+1/2}} \sigma_a(\nu, x, t) d\nu \quad (22)$$

$$\bar{B}_g(x, t) \equiv \int_{\nu_{i-1/2}}^{\nu_{i+1/2}} B(\nu, x, t) d\nu. \quad (23)$$

We assume that the opacity is Planck weighted before these definitions. From here we make the approximation

$$\int_{\nu_{i-1/2}}^{\nu_{i+1/2}} \sigma_a(\nu, t, x) \psi(\nu, t, x) d\nu = \bar{\sigma}_{\text{ag}} \bar{\psi}_g(t, x), \quad (24)$$

this allows for the neat formulation of the radiation equation:

$$\frac{1}{c} \frac{\partial}{\partial t} \bar{\psi}_g(t, x) + \Omega_x \left(\frac{\partial}{\partial x} \bar{\psi}_g(t, x) \right) = 4\pi\bar{\sigma}_{\text{ag}} \bar{B}_g - \bar{\sigma}_{\text{ag}} \bar{\psi}_g(t, x). \quad (25)$$

From here we move on to splitting the integral of the material equation

$$C_v \frac{\partial T(t)}{\partial t} = \sum_{g=1}^G \int_{\nu_{i-1/2}}^{\nu_{i+1/2}} (\sigma_a \phi(\nu, x, t) - 4\pi\sigma_a B) d\nu \quad (26)$$

$$C_v \frac{\partial T(t)}{\partial t} = \sum_{g=1}^G (\bar{\sigma}_{\text{ag}} \bar{\phi}(\nu, t, x) - 4\pi\bar{\sigma}_{\text{ag}} \bar{B}_g). \quad (27)$$

This leaves the full 1-d multigroup system. From here we again discretize the radiation and material equations, discretize angle, and apply the upwinding scheme. This leaves similar update equations to above. We once again define simplifying factors

$$Q^* = 4\pi\bar{\sigma}_{\text{ag}} \bar{B}_g - \bar{\sigma}_{\text{ag}} \bar{\psi}_g(t, x) \quad (28)$$

$$\sigma^* = \frac{1}{c\Delta t} + \bar{\sigma}_{\text{ag}}. \quad (29)$$

This makes the update equation

$$\bar{\psi}_{i,g}^l = \begin{cases} \frac{Q^* + \frac{\mu_l \bar{\psi}_{i-1}^l - 1}{\Delta x}}{\frac{\mu_l}{\Delta x} + \sigma^*}, & \text{for } \mu_l > 0 \\ \frac{Q^* + \frac{\mu_l \bar{\psi}_{i+1}^l - 1}{\Delta x}}{\frac{\mu_l}{\Delta x} + \sigma^*}, & \text{for } \mu_l < 0. \end{cases} \quad (30)$$

The material energy equation is coupled into Q^* , from here the

$$T_i^{n+1} = \frac{\Delta t}{C_v} \sum_{g=1}^G (\bar{\sigma}_{\text{ag}} \bar{\phi}_i^{n+1}(\nu, t, x) - 4\pi \bar{\sigma}_{\text{ag}} \bar{B}_g) + T_i^n. \quad (31)$$

3 Variable Basis Method

We can now define a new moving variable $u(x, t) = \frac{\nu}{T(x, t)}$. This allows the energy to evolve with the movement and development of the problem. We recall that the distribution of the new variable must match the old

$$\psi(\nu, x, t) d\nu = \hat{\psi}(u, t) du. \quad (32)$$

Taking the derivative of the new variable gives the relation between functions in frequency space and functions in the variable space

$$\phi(\nu, x, t) = \frac{\hat{\psi}(u, t)}{T(x, t)} \equiv \frac{F(u, t)}{T(x, t)}. \quad (33)$$

All functions in the new space will be delimited with hats. Subbing this into the radiation equation gives

$$\frac{1}{c} \frac{\partial}{\partial t} \frac{\hat{\psi}(u, t)}{T(x, t)} + \Omega_x \left(\frac{\partial}{\partial x} \frac{\hat{\psi}(u, t)}{T(x, t)} \right) = 4\pi \sigma_a \frac{\hat{B}(u, t)}{T(x, t)} - \sigma_a \frac{\hat{\psi}(u, t)}{T(x, t)}. \quad (34)$$

We can now simplify the individual terms

$$\frac{\partial}{\partial t} \frac{\hat{\psi}(u, t)}{T(x, t)} = \frac{1}{T(x, t)} \frac{\partial}{\partial t} \hat{\psi}(u, t) - \frac{\hat{\psi}(u, t)}{T^2} \frac{\partial T(x, t)}{\partial t}, \quad (35)$$

moving on to the spatial term

$$\frac{\partial}{\partial x} \frac{\hat{\psi}(u, t)}{T(x, t)} = \frac{1}{T(x, t)} \frac{\partial}{\partial x} \hat{\psi}(u, t) - \frac{\hat{\psi}(u, t)}{T^2} \frac{\partial T(x, t)}{\partial x}. \quad (36)$$

Because of the implicit time dependence in our variables, we must take the chain rule as follows

$$\frac{\partial}{\partial t} \hat{\psi}(u, t) = \frac{\partial}{\partial t} \hat{\psi}(u, t) + \frac{\partial}{\partial u} \hat{\psi}(u, t) \frac{\partial u}{\partial t} = \frac{\partial}{\partial t} \hat{\psi}(u, t) - \frac{u}{T(x, t)} \frac{\partial T(x, t)}{\partial t} \frac{\partial}{\partial u} \hat{\psi}(u, t) \quad (37)$$

$$\frac{\partial}{\partial x} \hat{\psi}(u, t) = \frac{\partial}{\partial x} \hat{\psi}(u, t) + \frac{\partial}{\partial u} \hat{\psi}(u, t) \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \hat{\psi}(u, t) - \frac{u}{T(x, t)} \frac{\partial T(x, t)}{\partial x} \frac{\partial}{\partial u} \hat{\psi}(u, t). \quad (38)$$

This makes the full first and second terms respectively

$$\frac{\partial}{\partial t} \frac{\hat{\psi}(u, t)}{T(x, t)} = \frac{1}{T(x, t)} \left(\frac{\partial}{\partial t} \hat{\psi}(u, t) - \frac{u}{T(x, t)} \frac{\partial T(x, t)}{\partial t} \frac{\partial}{\partial u} \hat{\psi}(u, t) \right) - \frac{\hat{\psi}(u, t)}{T^2} \frac{\partial T(x, t)}{\partial t} \quad (39)$$

$$\frac{\partial}{\partial x} \frac{\hat{\psi}(u, t)}{T(x, t)} = \frac{1}{T(x, t)} \left(\frac{\partial}{\partial x} \hat{\psi}(u, t) - \frac{u}{T(x, t)} \frac{\partial T(x, t)}{\partial x} \frac{\partial}{\partial u} \hat{\psi}(u, t) \right) - \frac{\hat{\psi}(u, t)}{T^2} \frac{\partial T(x, t)}{\partial x}. \quad (40)$$

The full radiation equation becomes

$$\begin{aligned} & \frac{1}{c} \left(\frac{1}{T(x, t)} \left(\frac{\partial \hat{\psi}(u, t)}{\partial t} - \frac{u}{T(x, t)} \frac{\partial T(x, t)}{\partial t} \frac{\partial \hat{\psi}(u, t)}{\partial u} \right) - \frac{\hat{\psi}(u, t)}{T^2} \frac{\partial T(x, t)}{\partial t} \right) \\ & + \Omega_x \left(\frac{1}{T(x, t)} \left(\frac{\partial \hat{\psi}(u, t)}{\partial x} - \frac{u}{T(x, t)} \frac{\partial T(x, t)}{\partial x} \frac{\partial \hat{\psi}(u, t)}{\partial u} \right) - \frac{\hat{\psi}(u, t)}{T^2} \frac{\partial T(x, t)}{\partial x} \right) = 4\pi \sigma_a \frac{\hat{B}(u, t)}{T(x, t)} - \sigma_a \frac{\hat{\psi}(u, t)}{T(x, t)}. \end{aligned} \quad (41)$$

We can simplify, canceling $T(x, t)$ from both sides, which leaves

$$\begin{aligned} & \frac{1}{c} \left(T(x, t) \frac{\partial \hat{\psi}(u, t)}{\partial t} - u \frac{\partial T(x, t)}{\partial t} \frac{\partial \hat{\psi}(u, t)}{\partial u} - \hat{\psi}(u, t) \frac{\partial T(x, t)}{\partial t} \right) \\ & + \Omega_x \left(T(x, t) \frac{\partial \hat{\psi}(u, t)}{\partial x} - u \frac{\partial T(x, t)}{\partial x} \frac{\partial \hat{\psi}(u, t)}{\partial u} - \hat{\psi}(u, t) \frac{\partial T(x, t)}{\partial x} \right) = 4\pi\sigma_a \hat{B}(u, t) - \sigma_a \hat{\psi}(u, t). \end{aligned} \quad (42)$$

Grouping terms gives

$$\begin{aligned} & T(x, t) \frac{\partial \hat{\psi}(u, t)}{\partial t} + c\Omega_x T(x, t) \frac{\partial \hat{\psi}(u, t)}{\partial x} - u \left(\frac{\partial T(x, t)}{\partial t} + c\Omega_x \frac{\partial T(x, t)}{\partial x} \right) \frac{\partial \hat{\psi}(u, t)}{\partial u} \\ & - \left(\frac{\partial T(x, t)}{\partial t} + c\Omega_x \frac{\partial T(x, t)}{\partial x} \right) \hat{\psi}(u, t) = 4\pi c\sigma_a \hat{B}(u, t) - c\sigma_a \hat{\psi}(u, t). \end{aligned} \quad (43)$$

We can define

$$\hat{V} = \left(\frac{\partial T(x, t)}{\partial t} + c\Omega_x \frac{\partial T(x, t)}{\partial x} \right), \quad (44)$$

Simplifying the overall radiation equation to

$$\begin{aligned} & T(x, t) \frac{\partial \hat{\psi}(u, t)}{\partial t} + c\Omega_x T(x, t) \frac{\partial \hat{\psi}(u, t)}{\partial x} - u \hat{V} \frac{\partial \hat{\psi}(u, t)}{\partial u} \\ & - \hat{V} \hat{\psi}(u, t) = 4\pi c\sigma_a \hat{B}(u, t) - c\sigma_a \hat{\psi}(u, t). \end{aligned} \quad (45)$$

This has the coupled material energy equation

$$C_v \frac{\partial T(t)}{\partial t} = \int_0^\infty d\nu (\sigma_a \hat{\phi}(u, t) - 4\pi\sigma_a \bar{B}_g). \quad (46)$$